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EN2150 - Communication Network Engineering

Routing Protocol Design

Team Lattice

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1 Abstract

This report presents Routing Information Protocol - Composite (RIP-C), an interior gateway protocol designed to overcome the hop-count limitations and slow convergence of legacy RIPv2. RIP-C introduces a normalized, bandwidth-and-latency-aware composite metric combined with authoritative Destination Sequence Numbers, shifting the architecture from a timer-based broadcast model to a purely event-driven framework. Empirical simulation results demonstrate that RIP-C achieves an **84.2%** reduction in convergence time, a **79.0%** optimization in average path latency, and a **101.0%** increase in average bottleneck bandwidth compared to RIPv2. While these mechanisms inherently eliminate routing loops and legacy hold-down timers, they introduce a 60% increase in routing entry payload size and require strict domain-wide synchronization of metric weights, rendering RIP-C highly effective for small-to-medium scale heterogeneous autonomous systems.

2 Simulation Code

The complete simulation source code and the guide are available in the project repository at <https://github.com/malindailankoon/EN2150-Routing-Protocol-Design-Simulation>.

3 Literature Review

3.1 RIP Protocol

The Routing Information Protocol (RIP) is an interior gateway protocol used to route computations within computer networks of moderate size. It is based on the Bellman-Ford, or distance vector, algorithm. Distance vector algorithms are based on the exchange of only a small amount of information. Each router or host participating in RIP simply keeps a routing database with one entry for every possible destination in the system, which includes the next gateway and a metric measuring the total distance to that entity.

3.1.1 Routing Table and Information Exchange

RIP routers periodically share their routing tables in their entirety. Every 30 seconds, the RIP process is awakened to send an unsolicited Response message containing the complete routing table to every neighboring router. RIP is a UDP-based protocol, and all routing update messages are sent from and received on UDP port number 520.

3.1.2 Neighbor Discovery and Adjacency Formation

RIP relies on listening to updates from neighbors on shared networks.

- In RIP version 1, unsolicited routing updates are normally sent as broadcasts on connected networks.
- In RIP version 2, an IP multicast address (224.0.0.9) is used for periodic broadcasts to reduce the unnecessary load on hosts that are not listening to RIP-2 messages.

3.1.3 Path Calculation: Distance Vector Algorithm

A RIP router relies on its neighbors' routing computations. When a router receives an update from a neighbor, it adds the cost associated with the network over which the update arrived to each of the destination metrics. If the newly calculated distance for a destination is smaller than the existing value in the router's table, it adopts the new route and notes the neighbor as the next hop. If an update arrives from the router currently used as the next hop for a specific route, the router will always update its metric to the new value, even if the new metric is larger.

3.1.4 Metric Definition: Hop Count

The primary metric used by RIP to compare alternative routes is a fixed integer between 1 and 15, inclusive.

- This metric represents the total cost of getting a datagram from the router to that destination, which is the sum of the costs associated with the traversed networks.
- Because the cost of a directly connected network is usually set to 1, the RIP metric typically reduces to a simple hop-count.
- A specific metric value of 16 is used to indicate an unreachable destination.
- This value of 16 is referred to as "infinity" because it is larger than the largest valid metric.

3.1.5 Loop Prevention

Because RIP relies on neighbors' tables rather than a complete map, it is vulnerable to routing loops and the "counting to infinity" problem, where vestiges of failed routes persist and metrics slowly increase back and forth between routers. RIP implements several specific mechanisms to prevent this instability:

- **Split Horizon:** The "simple split horizon" scheme omits routes learned from one neighbor in the routing updates sent back to that same neighbor.
- **Split Horizon with Poisoned Reverse:** This scheme includes the learned routes in updates sent back to the neighbor, but sets their metrics to infinity (16). This immediately breaks loops if two routers have routes pointing at each other.
- **Triggered Updates:** Whenever a router changes the metric for a route, it is required to send update messages almost immediately, rather than waiting for the regular 30-second timer. This cascades backwards along paths, speeding up convergence when a network becomes unreachable.

3.1.6 Hierarchy and Scalability

RIP was primarily intended for use as an interior gateway protocol in networks of moderate size. The protocol enforces a hard limit restricting usage to networks whose longest path (the network's diameter) is 15 hops. The designers believed the basic protocol design was inappropriate for larger networks.

3.1.7 Security and Authentication

The original RIP-1 protocol did not consider authentication. RIP-2 introduced an extensible authentication mechanism to improve security.

- If the Address Family Identifier of the first entry in a RIP-2 message is 0xFFFF, the remainder of that entry contains authentication data.
- The standard defines Authentication Type 2, which utilizes a simple plain text password up to 16 octets long.
- Additional cryptographic security improvements, such as MD5 authentication, were defined in companion documents to enhance RIP-2's security posture.

3.1.8 Known Limitations

- **Strict Size Limits:** The protocol is strictly limited to networks with a diameter of 15 hops or less. A hop count of 16 guarantees that a route is considered invalid and unreachable.
- **Bandwidth Overhead on Large Networks:** Broadcasting the entire routing table every 30 seconds can consume a large amount of network bandwidth before loops are corrected, particularly on slow lines. While split horizon with poisoned reverse improves

dynamic behavior, it can significantly increase the size of update messages, resulting in wasted bandwidth from entries indicating unreachable networks.

- **Slow Convergence:** In the event of a topology change where a network becomes inaccessible, the system relies on wait timers (such as waiting 180 seconds before timing out a route) and can suffer from slow convergence while it "counts to infinity".
- **Incomplete Cost Design:** RIP uses fixed metrics to compare alternative routes. It is not appropriate for situations where routing decisions need to be based on real-time parameters such as measured delay, reliability, or network load. Extensions allowing such metrics are likely to introduce instability.

3.2 OSPF Protocol

OSPF is a link-state interior gateway protocol used to route packets within a single autonomous system. Unlike distance-vector protocols that share routing tables, OSPF routers share a complete map of the network topology, then each calculates its own best paths independently.

3.2.1 LSA Architecture and Information Exchange

OSPF routers don't share routing tables. Instead, they flood Link-State Advertisements (LSAs) across the network. Each router describes its own directly connected links; their state, cost, and neighbors using LSAs. Every router collects all LSAs and builds an identical Link-State Database (LSDB), which is essentially a complete map of the network. From this shared map, each router independently calculates its own best routes. There are several LSA types. The most common are Type 1 (Router LSA) describing a router's own links, Type 2 (Network LSA) describing multi-access network segments, and Type 3 (Summary LSA) carrying route summaries between OSPF areas.

3.2.2 Neighbor Discovery and Adjacency Formation

OSPF uses Hello packets sent as multicast to discover neighbors on a link. Before any LSAs are exchanged, two routers must form an adjacency by progressing through a state machine.

- **Down → Init:** Hello received from neighbor
- **Init → 2-Way:** Own Router ID seen in neighbor's Hello (bidirectional communication confirmed)
- **2-Way → ExStart:** Begin deciding who is Master/Slave for database exchange

- **ExStart → Exchange:** DBD (Database Description) packets summarise each router's LSDB
- **Exchange → Loading:** LSR (Link-State Request) sent for missing entries; LSU (Link-State Update) received in reply
- **Loading → Full:** LSDB is synchronised — adjacency is fully established

3.2.3 Path Calculation: Dijkstra's SPF Algorithm

Once the LSDB is complete, each router independently runs Dijkstra's Shortest Path First (SPF) algorithm. The router treats itself as the root of a tree and calculates the shortest (lowest-cost) path to every other destination. This produces the SPF tree, and the best next-hops are installed into the routing table. The algorithm works in stages: start with the local router at cost 0, examine all neighbors, add the cheapest unvisited node to the "confirmed" set, and repeat until all nodes are reached.

3.2.4 Metric Definition: Cost

$$Cost = \frac{\text{Reference Bandwidth}}{\text{Interface Bandwidth}}$$

By default, the reference bandwidth is 1000 Mbps.

Interface	Bandwidth	OSPF Cost
10 Mbps (Ethernet)	10 Mbps	100
100 Mbps (Fast Ethernet)	100 Mbps	10
1 Gbps (GigE)	1000 Mbps	1
10 Gbps	10,000 Mbps	1

3.2.5 Loop Prevention

OSPF is inherently loop-free because of its link-state nature. Since every router holds an identical complete map of the topology, and Dijkstra's SPF produces a loop-free tree rooted at each router, routing loops cannot form. There is no possibility of the "counting to infinity" problem present in distance-vector protocols. Additionally, OSPF uses sequence numbers on LSAs so outdated or duplicate LSAs are discarded. An age field causes LSAs to expire after a maximum age, preventing stale information from persisting indefinitely.

3.2.6 Hierarchy and Scalability: Areas

OSPF scales through a two-level hierarchy of areas. All areas must connect to Area 0 (the backbone area). Breaking a large network into multiple areas limits LSA flooding and SPF recalculations to within each area, dramatically improving scalability.

3.2.7 Security and Authentication

OSPF supports three authentication modes, configured per-interface or per-area:

- **Null (Type 0) No authentication:** the default. Any router sending OSPF packets will be accepted. Suitable only for isolated lab environments.
- **Simple/Clear-text (Type 1):** A plain-text password is included in OSPF packet headers. Protects against accidental misconfiguration but offers no real security. the password is visible in a packet capture.
- **MD5 (Type 2):** A cryptographic hash (MD5) of the packet contents plus a shared secret key is appended to each packet. The receiver independently computes the hash and rejects the packet if it does not match. This prevents both passive eavesdropping and active injection of forged LSAs. Multiple keys with different key IDs can be configured simultaneously to allow graceful key rollover without breaking adjacencies.

Modern deployments increasingly use OSPFv3 with IPsec (for IPv6 networks), or SHA-HMAC authentication extensions, which address known weaknesses in MD5.

3.2.8 Known Limitations

- **Complexity of Configuration and Management:** OSPF is significantly more complex to configure than simpler protocols like RIP or EIGRP. Area design, ABR placement, and route summarization require careful planning and are difficult to debug on large networks.
- **Memory and CPU Demands:** Every router must store the complete LSDB for every area it belongs to. In large areas with many routers and links, this database can grow very large, consuming significant RAM. SPF recalculations triggered every time a topology change occurs are also CPU-intensive.
- **Incomplete Cost Design:** OSPF's cost metric only considers bandwidth. It has no native awareness of:
 - **Latency:** A high-bandwidth but high-delay satellite link looks identical to a fast local link if bandwidths match.

- **Load or Utilisation:** A congested link is treated the same as an idle one.
- **No Native Load Balancing Across Unequal Paths:** OSPF supports Equal-Cost Multi-Path (ECMP) load balancing. It can spread traffic across multiple paths, but only if they have identical cost. Unlike EIGRP, which supports unequal-cost load balancing, OSPF will simply ignore a path the moment it has a higher cost than the best path, even if that path is nearly as good and has spare capacity. Getting equal costs across diverse paths often requires manual cost manipulation.

3.3 IS-IS Protocol

IS-IS (Intermediate System to Intermediate System) is a link-state interior gateway routing protocol mainly used in large enterprise and ISP backbone networks. Similar to OSPF, IS-IS distributes topology information throughout the network so that every router can maintain a complete view of the topology and independently calculate shortest paths using Dijkstra's Shortest Path First (SPF) algorithm.

Originally developed for the OSI protocol suite, IS-IS was later extended to support IP routing and is now widely deployed in large-scale service provider networks because of its scalability, flexibility, and stable hierarchical design.

3.3.1 Information Exchange

IS-IS exchanges routing information using Link-State Packets (LSPs). Each router generates LSPs containing information about its connected links and neighboring routers, then floods them throughout the routing domain.

- IS-IS uses a flexible TLV (Type-Length-Value) structure, allowing new features and extensions to be added without redesigning the protocol.
- All routers store received LSPs in a Link-State Database (LSDB), enabling every router to maintain a synchronized network topology map.
- Efficient flooding mechanisms help distribute topology changes rapidly across the network.
- However, excessive LSP flooding in very large or unstable networks can increase bandwidth consumption and processing overhead.

3.3.2 Neighbor Discovery and Adjacency Formation

IS-IS routers discover neighboring routers using Hello packets, commonly referred to as IS-IS Hellos (IIHs). Routers exchange these packets periodically to establish and maintain

adjacencies.

- Adjacencies are formed only when routers agree on important parameters such as area ID and authentication settings.
- Stable adjacency formation ensures reliable synchronization of routing information between routers.
- IS-IS supports operation over both point-to-point and broadcast network types.
- Incorrect configuration parameters may prevent adjacency establishment and disrupt routing communication.

3.3.3 Path Calculation: Dijkstra's SPF Algorithm

After the LSDB is synchronized, each router independently executes Dijkstra's SPF algorithm to determine the shortest path to every destination.

- Because all routers maintain an identical topology database, routing decisions remain highly consistent across the network.
- SPF calculations produce loop-free shortest-path trees rooted at each router.
- IS-IS generally provides faster and more stable path calculation compared to traditional distance-vector protocols.
- However, repeated SPF recalculations caused by frequent topology changes can increase CPU utilization on routers.

3.3.4 Metric Definition

IS-IS uses configurable cost metrics to determine the best path between destinations.

- Network administrators can assign interface costs based on bandwidth, delay, or specific routing policies.
- Lower cumulative path costs are preferred during route calculation.
- Flexible metric configuration allows better traffic engineering and network optimization.
- Improper metric assignments may lead to inefficient routing decisions or uneven traffic distribution.

3.3.5 Loop Prevention

IS-IS minimizes routing loops through its link-state architecture and synchronized topology database.

- Since all routers calculate routes from the same LSDB, routing inconsistencies are significantly reduced.
- Sequence numbers are attached to LSPs so routers can identify and discard outdated routing information.
- Aging mechanisms automatically remove stale entries from the LSDB.
- Temporary routing loops may still occur briefly during rapid topology transitions or convergence events.

3.3.6 Convergence Speed

IS-IS provides relatively fast convergence because topology changes are propagated immediately using LSP flooding.

- Routers can quickly update their LSDBs after link or node failures.
- Fast convergence improves network stability and reduces packet loss during failures.
- IS-IS generally converges faster than traditional distance-vector protocols such as RIP.
- However, convergence performance can degrade in highly unstable or extremely large-scale networks due to excessive SPF recalculations.

3.3.7 Hierarchy and Scalability

IS-IS is designed with hierarchical routing support, making it highly scalable for large networks.

- The protocol divides routing domains into Level-1 and Level-2 areas.
- Level-1 routers handle intra-area routing, while Level-2 routers connect multiple areas together.
- This hierarchical structure reduces routing overhead and improves scalability in large enterprise and ISP backbone networks.
- Despite its scalability advantages, maintaining very large LSDBs can increase memory and processing requirements on routers.

3.3.8 Security and Authentication

IS-IS supports authentication mechanisms to validate routing updates and prevent unauthorized routers from participating in the routing domain.

- Authentication improves routing security by reducing the risk of false route advertisements.
- Password-based authentication methods can be configured between neighboring routers.
- Modern implementations support stronger cryptographic authentication mechanisms for improved protection.
- Older authentication methods may still be vulnerable to attacks such as spoofing or denial-of-service (DoS) attacks if not properly secured.

3.3.9 Known Limitations

- **Configuration Complexity:** IS-IS configuration and troubleshooting can be complex, especially in large hierarchical networks.
- **High Resource Usage:** Large LSDBs and repeated SPF calculations may consume considerable CPU and memory resources.
- **Protocol Familiarity:** IS-IS is less commonly taught and deployed in enterprise environments compared to OSPF, making experienced administrators less widely available.
- **Scalability Overhead:** In extremely large or unstable networks, excessive LSP flooding and SPF recalculations can reduce overall routing efficiency.

3.4 BGP Protocol

BGP-4 is the global Internet's core routing protocol, enabling Classless Inter-Domain Routing (CIDR) to exchange reachability information. As a path-vector protocol, BGP routes traffic based on administrative policies and commercial agreements rather than shortest topological paths.

3.4.1 Information Exchange

BGP establishes explicitly configured TCP connections over port 179 for information exchange, safely offloading fragmentation and reliability to the transport layer. Networks exchange Network Layer Reachability Information (NLRI) using this reliable transport once a session is active.

3.4.2 Neighbor Discovery and Adjacency Formation

Unlike interior protocols, BGP does not use dynamic neighbor discovery; it relies on explicit peer configurations and a strict Finite State Machine (FSM). The FSM transitions through Idle, Connect, Active, OpenSent, OpenConfirm, and Established states. In the event of simultaneous connection attempts, the FSM gracefully resolves collisions by preserving the session initiated by the router with the numerically higher BGP Identifier.

3.4.3 Path Calculation

BGP path calculation relies on a deterministic sequential algorithm that evaluates path attributes to find the best route. It does not use a single mathematical cost. Instead, the algorithm evaluates attributes sequentially, terminating at the first tie-breaker, which ensures administrative traffic engineering overrides proximity.

3.4.4 Metric Definition

BGP evaluates path attributes in descending order of preference to define its metric routing logic. Highest preference is given to Weight (local router policy) and Local Preference (AS-wide outbound paths). It then evaluates AS_PATH Length, preferring the shortest inter-domain metric. Finally, it considers the Multi-Exit Discriminator (MED), which has the lowest preference and influences inbound traffic from adjacent ASes.

3.4.5 Loop prevention

BGP employs bifurcated loop prevention mechanisms depending on the context.

- For External BGP (eBGP), a router inspects the AS_PATH attribute and discards updates if it detects its own Autonomous System Number (ASN).
- For Internal BGP (iBGP), a router prevents loops using the split-horizon rule, which strictly prohibits routes learned from one iBGP peer from being advertised to another.

3.4.6 Hierarchy and Stability

The iBGP split-horizon rule requires a full logical mesh, creating severe scalability challenges. Networks resolve this using Route Reflectors (which centralize route distribution) or Confederations (which divide a large AS into sub-ASes). For stability and fast convergence, modern networks use Bidirectional Forwarding Detection (BFD) for rapid link failure detection, BGP Add-Path for instant failover, and Route Refresh for non-disruptive policy updates.

3.4.7 Security and Authentication

- BGP natively lacks cryptographic validation, leaving it highly vulnerable to route leaks and malicious prefix hijacking.
- Current mitigation efforts rely on the Resource Public Key Infrastructure (RPKI) to cryptographically validate Route Origin Authorizations (ROAs).
- Emerging standards like the Autonomous System Provider Authorization (ASPA) aim to enforce cryptographic path verification to stop advanced forged-origin attacks.

3.4.8 Known Limitations

- BGP is inherently slow to converge during topology changes without modern optimizations.
- The iBGP full-mesh requirement creates significant scaling limitations in large network cores.
- A critical modern limitation is that current anomaly detection techniques struggle with advanced, monitor-evasive attacks that strategically exploit blind spots in public routing infrastructure.
- Balancing the computational overhead of advanced routing architectures with the critical need for global cryptographic security adoption remains an ongoing challenge.

4 Protocol Design

4.1 Protocol Overview and Objectives

To address the critical limitations of standard RIP, namely its **incomplete cost design** and **slow convergence** due to the "counting to infinity" problem, we propose **RIP-C (Routing Information Protocol - Composite)**. RIP-C remains an Interior Gateway Protocol (IGP) utilizing a distance-vector algorithm, but it is enhanced with a bandwidth and latency aware composite metric and a sequence based loop prevention mechanism.

4.2 The Composite Metric Design

Standard RIP relies on a simplistic hop-count metric, which fails to distinguish between high-capacity fiber links and congested, slow connections. RIP-C replaces this with a composite metric combining the path's bottleneck bandwidth and cumulative static latency.

To ensure both metrics scale appropriately without latency being mathematically overshadowed on slow links or vice versa, RIP-C normalizes the units. It utilizes a **100 Gbps reference bandwidth** (expressed as 100,000 Mbps) and measures latency in **microseconds** (μs).

The final route cost is calculated using the following formula:

$$Cost = \left(K_1 \times \frac{100,000}{\text{Bottleneck Bandwidth (Mbps)}} \right) + (K_2 \times \text{Cumulative Latency } (\mu s)) \quad (1)$$

Where K_1 and K_2 are **configurable positive weighting constants** (integers), allowing network administrators to prioritize either throughput or delay based on network requirements. This design allows RIP-C to be deployed in subnetworks that contain **dissimilar link capacities**.

4.3 Message Format

Standard RIPv2 utilizes a 20-byte route entry format. To accommodate the new composite metric variables and loop-prevention mechanisms, RIP-C expands the route entry size to **32 bytes**.

Byte Offset	Length	Field Name	Description
0	2 Bytes	Address Family	Set to 2 for IPv4.
2	2 Bytes	Route Tag	Used to tag external routes (kept for legacy support).
4	4 Bytes	IP Address	The destination network address.
8	4 Bytes	Subnet Mask	The subnet mask for the destination.
12	4 Bytes	Next Hop	The IP address of the next-hop router.
16	4 Bytes	Bottleneck Bandwidth	The lowest bandwidth on the path, measured in Mbps.
20	4 Bytes	Cumulative Latency	The total path delay, measured in microseconds (μs).
24	4 Bytes	Sequence Number	Used for mathematical loop prevention.
28	4 Bytes	Metric (Cost)	The pre-calculated integer cost. Used for rapid comparison assuming K_1 and K_2 weights are standardized across the system.

Table 1: RIP-C 32-Byte Route Entry Format

4.4 Router State

Each router participating in RIP-C must maintain an expanded routing table to support the composite algorithm. For every destination network, the routing database stores:

- Destination Network Address and Subnet Mask
- Next-Hop Gateway IP Address
- Path Bottleneck Bandwidth (Mbps)
- Path Cumulative Latency (μs)
- Destination Sequence Number: The highest known sequence number generated for the destination node.
- Route State (Active / Invalid)

4.5 Operation Rules and Algorithms

RIP-C modifies the standard Bellman-Ford algorithm to utilize sequence numbers, **providing instantaneous loop prevention** and rendering standard RIP's 30-second timers and poisoned-reverse mechanics obsolete.

When a router receives an update from a neighbor, it evaluates the route based on the following logic:

1. **Update Variables:** The router **adds** the receiving interface's static delay to the advertised Cumulative Latency. It compares the receiving interface's bandwidth against the advertised Bottleneck Bandwidth, retaining the **lower of the two values**.
2. **Calculate Metric:** The router calculates the new Total Cost using the composite formula.
3. **Sequence Verification (Loop Prevention):**
 - **Rule 1 (Timeliness):** If the incoming update contains a **higher Sequence Number** for the destination than the currently stored route, the router **must** accept the new route immediately, regardless of the cost metric. This breaks "counting to infinity" loops instantly.
 - **Rule 2 (Optimization):** If the incoming Sequence Number is **equal** to the stored sequence number, the router adopts the route **only** if the newly calculated Cost is **lower** than the existing stored Cost.

Listing 1: RIP-C Route Evaluation Algorithm

```

1 BEGIN Process_Update(Incoming_Packet, Receiving_Interface)
2   Dest = Incoming_Packet.Destination
3   Adv_Seq = Incoming_Packet.Sequence_Number
4   Adv_BW = Incoming_Packet.Bottleneck_Bandwidth
5   Adv_Lat = Incoming_Packet.Cumulative_Latency
6
7   Local_Lat = Adv_Lat + Receiving_Interface.Static_Delay
8   Local_BW = MIN(Adv_BW, Receiving_Interface.Bandwidth)
9   New_Cost = (K1 * (100000 / Local_BW)) + (K2 * Local_Lat)
10
11  IF Dest NOT IN Routing_Table THEN
12    IF New_Cost < INFINITY THEN
13      Add_Route(Dest, NextHop=Incoming_Packet.Sender,
14                Cost=New_Cost, Seq=Adv_Seq)
15      Trigger_Flash_Update()
16    END IF
17  ELSE
18    Current_Seq = Routing_Table[Dest].Sequence
19    Current_Cost = Routing_Table[Dest].Cost
20
21    IF Adv_Seq > Current_Seq THEN
22      Update_Route(Dest, NextHop, New_Cost, Adv_Seq)
23      Trigger_Flash_Update()
24
25    ELSE IF Adv_Seq == Current_Seq THEN
26      IF New_Cost < Current_Cost THEN
27        Update_Route(Dest, NextHop, New_Cost, Adv_Seq)
28        Trigger_Flash_Update()

```

```

29         ELSE
30             Drop_Packet()
31         END IF
32
33     ELSE
34         Drop_Packet()
35     END IF
36 END IF
37 END Process_Update

```



Figure 1: RIP-C Route Evaluation Flowchart

Instead of periodic 30-second broadcasts, RIP-C relies entirely on **Triggered Updates**. A router will only broadcast its routing table if a link state changes, a sequence number increments, or a metric shift occurs, reducing idle bandwidth consumption.

4.6 Loop Prevention Mechanisms

Standard distance-vector protocols suffer from the "counting to infinity" problem. When a network becomes unreachable, routers relying on neighbors' routing tables can create transient routing loops, incrementally updating metrics until they reach the designated infinity value. Legacy RIP mitigates this using Split Horizon and Poisoned Reverse, which add complexity and scale poorly.

RIP-C eliminates this vulnerability inherently through the implementation of **Destination Sequence Numbers**.

- **Authoritative Stamping:** During normal operation, **only** the destination router is permitted to generate or increment the sequence number for its directly connected networks.
- **Failure Detection and Proxy Invalidation:** If the destination router's metric state changes, it increments its own sequence number. If the destination suffers a total failure, RIP-C exceptionally permits immediate neighbors to increment the sequence number to broadcast an unreachable state (detailed in Section 4.8).
- **Instant Override:** Because RIP-C enforces the rule that a higher sequence number **always** supersedes existing routing data, neighboring routers immediately accept the new "infinity" metric. The stale path information is instantly flushed from the network, preventing any loops from occurring.

4.7 Convergence Optimization

Standard RIP dictates a 30-second periodic update interval and utilizes a 180-second hold-down timer before invalidating a non-responsive route, leading to high convergence times during topology changes.

RIP-C reduces convergence time by abandoning periodic broadcast timers in favor of an event driven model:

- **Triggered Updates:** RIP-C routers do not broadcast their routing tables on a fixed schedule. Updates are exclusively triggered by topology changes, such as a link failure, a link restoration, or a metric change.

- **Removal of Hold-Down Timers:** Because sequence numbers guarantee loop-free topologies, the legacy hold-down timers used to prevent routing loops are rendered obsolete. Routers can immediately adopt new alternative paths without waiting for timeout periods, reducing network downtime.

4.8 Handling Edge Cases

4.8.1 Cold Start and Network Initialization

Because RIP-C eliminates periodic broadcasts, a newly added router cannot passively wait to learn the network topology. RIP-C resolves this using a "Cold Start" request mechanism:

- **Initialization Request:** Upon booting or enabling a RIP-C process on an interface, the router immediately multicasts a Request Packet. This packet serves as an active query to all attached neighbors, indicating a lack of topological state.
- **Unicast Synchronization:** Receiving neighbors treat this Request as a triggered event. They bypass normal update rules and immediately unicast their complete, current routing table back to the requesting router.
- **Link-Up Trigger:** Simultaneously, the existing network detects the new router's interface transitioning to an "Up" state. This topology change naturally acts as a triggered update, causing neighbors to propagate the new link's availability throughout the system.

4.8.2 Authoritative Stamping and Proxy Invalidation

To maintain routing integrity while ensuring fault tolerance, RIP-C enforces a dual-layered sequence number rule:

- **Active State Generation:** During normal operation, only the destination router is authorized to generate and increment its sequence number to advertise active, reachable paths. It increments this counter **only** when a local topology change occurs.
- **Proxy Invalidation:** If a destination router suffers a hardware or link failure, it cannot broadcast an updated state. Therefore, RIP-C authorizes immediate adjacent neighbors who detect the physical link failure to act as proxies. The neighbor is permitted to increment the destination's last known sequence number by 1, provided it simultaneously sets the composite cost to Infinity.

Listing 2: RIP-C Handling a Link Failure

```
1 BEGIN Handle_Link_Failure(Failed_Interface)
2   FOR EACH Target IN Failed_Interface.Connected_Networks
3     Current_Seq = Routing_Table[Target].Sequence
4
5     Routing_Table[Target].Sequence = Current_Seq + 1
6     Routing_Table[Target].Cost = INFINITY
7     Routing_Table[Target].State = INVALID
8
9     Trigger_Flash_Update()
10  END FOR
11 END Handle_Link_Failure
```

- **Redundant Detection Resolution:** In highly connected topologies where multiple routers are directly adjacent to the failed node, all adjacent routers will independently detect the failure, increment the sequence number, and advertise an Infinity cost. Because all neighbors derive the identical incremented sequence number and Infinity metric, the network processes these as non-conflicting updates.

4.8.3 Routing Race Condition Resolution

Because updates propagate at varying speeds across different paths, RIP-C may encounter a routing race condition where an update from a suboptimal path arrives before an update from the optimal path.

1. If a router receives a new sequence number via a slow, high-cost path first, Rule 1 dictates it must accept the route immediately to ensure timeliness. For a brief period, traffic may route sub-optimally.
2. Sometime later, the update carrying the identical sequence number arrives via the faster, low-cost path.
3. Because the sequence numbers match, Rule 2 is triggered. The router compares the composite metrics, drops the suboptimal path, and stores the low-cost path.

4.8.4 Sequence Number Wraparound

The 32-bit sequence number utilizes circular arithmetic. If the mathematical difference between a newly received sequence number and the locally stored sequence number is negative, but exceeds half the maximum 32-bit value (2^{31}), the protocol recognizes that a wraparound has occurred. The router correctly identifies the numerically lower value as the fresher routing update, ensuring continuous operation.

4.9 Known Limitations and Trade-offs

4.9.1 Domain-Wide Parameter Synchronization

The flexibility of the composite metric relies entirely on the synchronization of the weighting constants (K_1 and K_2) across the routing domain. If human error results in an administrative misconfiguration where neighboring routers possess varying K values, the routers will calculate asymmetric path costs. This desynchronization breaks the Bellman-Ford optimization logic and can introduce artificial routing loops. RIP-C lacks a native pre-validation handshake to enforce K -value matching prior to adjacency formation.

4.9.2 Absence of Hierarchical Abstraction

Despite resolving the 15-hop limit and the "counting to infinity" problem, RIP-C remains as a Interior Gateway Protocol. Unlike OSPF or IS-IS, which utilize hierarchical boundaries (Areas) to confine topology changes to localized segments, a triggered update in RIP-C must propagate through the entire routing domain to converge. Therefore, RIP-C is unsuitable for massive enterprise or service-provider backbones where route summarization and fault isolation are mandatory.

4.9.3 Increased Payload Overhead

To accommodate the transmission of Bottleneck Bandwidth, Cumulative Latency, and 32-bit Sequence Numbers, the RIP-C route entry structure was expanded from 20 bytes to 32 bytes. This 60% increase in payload overhead means more bandwidth utilization is expected during initialization or during a link failure, compared to RIP.

5 Performance Analysis

The performance of the proposed RIP-C protocol is evaluated and compared against the legacy RIPv2 protocol across four critical metrics: **Convergence Time**, the **Number of Packets** required to reach convergence, **Average Latency**, and **Average Bottleneck Bandwidth**. The specific network topology utilized for this analysis is illustrated in Figure 2.

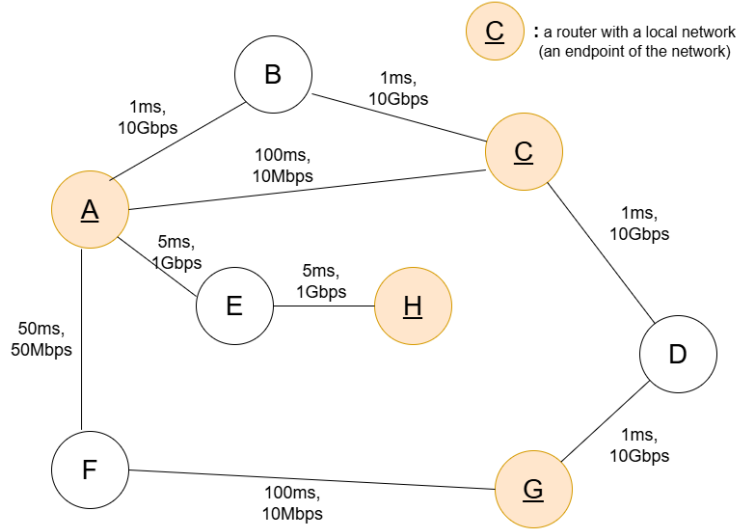


Figure 2: Simulation Test Network Topology

The average end-to-end path latency across the converged network is determined using the following equation:

$$L_{avg} = \frac{\sum_{i=1}^n L_i}{n}$$

Where:

- L_{avg} represents the average network latency, measured in milliseconds (ms).
- n denotes the total number of active, reachable remote routing paths across all operational nodes in the network topology.
- L_i represents the cumulative propagation latency of the i -th individual path, calculated by summing the static propagation delays (d) of each traversed physical link along that specific route such that $L_i = \sum d$.

The average bottleneck bandwidth across the network infrastructure is calculated using the expression:

$$B_{avg} = \frac{\sum_{i=1}^n \min(b_{i,1}, b_{i,2}, \dots, b_{i,m})}{n}$$

Where:

- B_{avg} represents the average path capacity across the active network topology, expressed in Megabits per second (*Mbps*).
- n denotes the total number of active, valid remote routing paths within the network core.
- $\min(b_{i,1}, b_{i,2}, \dots, b_{i,m})$ represents the bottleneck bandwidth of the i -th active path, determined by isolating the minimum link capacity (b) among all m individual physical segments constituting that end-to-end route.

5.1 Simulation Results and Evaluation

```
--results for StandardRIPv2 --
Total Packets Sent   : 967
Routing Updates      : 29
Convergence Time     : 5.02 s
Average Latency      : 61.50 ms
Average Bandwidth    : 2340.36 Mbps
Total Active Paths   : 28

--results for RIP_C --
Total Packets Sent   : 93
Routing Updates      : 30
Convergence Time     : 0.79 s
Average Latency      : 12.86 ms
Average Bandwidth    : 4721.43 Mbps
Total Active Paths   : 28
```

Figure 3: Comparative Analysis of Initial Boot Convergence Metrics

Based on the data captured from the simulation environment and shown in Figure 3, transitioning from standard timer based RIPv2 to the proposed event driven RIP-C architecture yields the following performance enhancements:

- An **84.2% reduction** in network **Convergence Time**.
- A **79.0% reduction** in the **Average Latency** of established routing paths.
- A **101.0% increase** in the **Average Bottleneck Bandwidth** across active routes.

Furthermore, the total volume of control plane routing updates transmitted to achieve steady-state convergence is significantly lower in RIP-C. This is a direct consequence of shifting from a periodic broadcast scheme to an event driven triggered flash update model.

A secondary simulation experiment was executed to evaluate the dynamic resilience of the network under unexpected structural changes, specifically focusing on the physical failure of the link between Node A and Node C. As illustrated in Figure 4, RIP-C demonstrates the performance advantage over legacy RIPv2 when managing link-state variations.

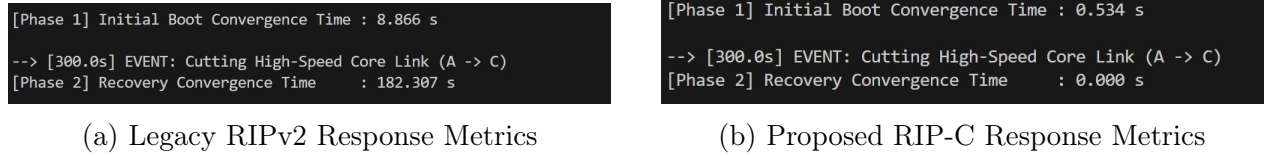


Figure 4: Simulation Results Evaluating Network Rerouting and Convergence Following a Structural Link Failure

6 Security and Scalability Analysis

6.1 Security

Cryptographically, Standard RIPv2 and RIP-C share an identical baseline, utilizing legacy plaintext authentication (AFI 0xFFFF) rather than robust hashing mechanisms like MD5. However, their architectural resilience and vulnerability surfaces differ.

6.1.1 Resilience Against Routing DoS (RIP-C Advantage)

Legacy RIPv2 is vulnerable to network-degrading anomalies, particularly the “counting to infinity” problem. Malicious route injections or rapidly flapping links can trigger endless routing loops, acting as Denial of Service (DoS) attacks that exhaust bandwidth and CPU cycles.

RIP-C eliminates this via **Authoritative Stamping**. Because only the true destination router can increment its sequence number, stale or maliciously injected routes are suppressed. A rogue router cannot easily hijack a prefix, as the legitimate destination’s continuously incrementing sequence number will mathematically override forged advertisements across the domain.

6.1.2 The K -Value Exploit (RIP-C Vulnerability)

While resolving loop vulnerabilities, RIP-C introduces a new attack surface: **Domain-Wide Parameter Synchronization**. RIP-C assumes the weighting constants (K_1 and K_2) are identical across all nodes but lacks a pre-validation handshake to enforce this during adjacency formation.

If an internal attacker or administrative misconfiguration alters these K values on a single router, it will calculate asymmetric path costs. This desynchronization breaks the Bellman-Ford optimization logic, allowing bad actors to silently blackhole traffic or purposefully induce routing loops. Implementing a K -value validation handshake during initialization is a mandatory prerequisite for secure deployment.

6.2 Scalability

RIP-C trades payload efficiency for convergence speed. By expanding the route entry from 20 bytes to 32 bytes, RIP-C consumes more bandwidth during initial "Cold Start" and after link failures. Furthermore, because RIP-C remains a flat distance-vector protocol without hierarchical abstractions, every triggered flash update must propagate across the entire routing domain. Therefore, while RIP-C outperforms legacy RIP in small-to-medium heterogeneous networks, its flat architecture and increased payload size **limit its scalability** for massive global enterprise deployments.

7 Conclusion

In conclusion, the design and implementation of RIP-C successfully demonstrates that integrating a composite cost function with Destination Sequence Numbers effectively modernizes traditional distance-vector routing. By transitioning to an event driven update model, the protocol eliminates routing loops and the "counting to infinity" vulnerability at line rate, making legacy loop prevention workarounds completely obsolete. Performance metrics validated the theoretical objectives, showing substantial improvements in convergence speed, link capacity utilization, and latency optimization under both initial boot and active link-failure states. Future iterations will focus on mitigating the identified trade-offs, specifically addressing the 32-byte packet overhead and introducing architectural handshakes to prevent performance degradation caused by desynchronized administrative weights (K_1 and K_2).

8 Individual Contributions

- Jayasinghe J.A.P.R. (230280L)

- **In Charge:** Literature Review
- **Literature Review:** 70% contribution
- **Protocol Design:** 10% contribution
- **Simulation:** 10% contribution
- **Security & Scalability Analysis:** 0% contribution
- **Samarasinghe S.M.R.R. (230566U)**
 - **In Charge:** Protocol Design
 - **Literature Review:** 10% contribution
 - **Protocol Design:** 70% contribution
 - **Simulation:** 10% contribution
 - **Security & Scalability Analysis:** 0% contribution
- **Ilankoon I.M.M.K.B. (230256U)**
 - **In Charge:** Simulation
 - **Literature Review:** 10% contribution
 - **Protocol Design:** 10% contribution
 - **Simulation:** 45% contribution
 - **Security & Scalability Analysis:** 30% contribution
- **Imaduwege O.N.H. (230258D)**
 - **In Charge:** Security & Scalability Analysis
 - **Literature Review:** 10% contribution
 - **Protocol Design:** 10% contribution
 - **Simulation:** 35% contribution
 - **Security & Scalability Analysis:** 70% contribution

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